EXPERIMENT - 11

Step 1: Understand the Problem Statement

Problem: The user enters a positive integer and then chooses one of three operations from a menu. Depending on the choice, the program extracts the digits of the number and computes the required result using the specific loop mentioned for that operation.

Required Input: a positive integer (num), and the menu choice (1, 2, or 3)

Required Output: sum of squares of digits (choice 1), sum of cubes of digits (choice 2), or the difference between sum of cubes and sum of squares (choice 3)

Formulas Used:

- Choice 1 (while loop): sumSquares = Σ (digit × digit) for every digit of num
- Choice 2 (for loop): sumCubes = Σ (digit × digit × digit) for every digit of num
- Choice 3 (do-while loop): difference = sumCubes − sumSquares

Step 2: Prepare the Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	num = 123 choice = 1	Sum of squares of digits = 14	Sum of squares of digits = 14	**Pass**
2	num = 123 choice = 2	Sum of cubes of digits = 36	Sum of cubes of digits = 36	**Pass**
3	num = 123 choice = 3	Difference (Sum of cubes - Sum of squares) = 22	Difference (Sum of cubes - Sum of squares) = 22	**Pass**
4	num = 9 choice = 1	Sum of squares of digits = 81	Sum of squares of digits = 81	**Pass**
5	num = 45 choice = 3	Difference (Sum of cubes - Sum of squares) = 148	Difference (Sum of cubes - Sum of squares) = 148	**Pass**
6	num = 123 choice = 9	Invalid choice!	Invalid choice!	**Pass**

Step 3: Algorithm

1. Start

2. Read a positive integer num

3. Display the menu with 3 choices and read the user's choice

4. If choice = 1: initialize sumSquares = 0; using a while loop, repeatedly extract the last digit of num (digit = num % 10), add digit × digit to sumSquares, and remove the last digit (num = num / 10) until num becomes 0; then display sumSquares

5. If choice = 2: initialize sumCubes = 0; using a for loop, repeatedly extract the last digit of num, add digit × digit × digit to sumCubes, and remove the last digit until num becomes 0; then display sumCubes

6. If choice = 3: initialize sumSquares = 0 and sumCubes = 0; using a do-while loop, extract each digit and accumulate both sumSquares and sumCubes at least once before checking the loop condition; compute difference = sumCubes − sumSquares and display it

7. If choice is none of the above, display "Invalid choice!"

8. Stop

Evaluation against test cases: Tracing the algorithm manually for each test case in Step 2 (e.g., for num = 123 and choice = 1: digits 1, 2, 3 → sumSquares = 1 + 4 + 9 = 14) produces results that exactly match the expected output for all 6 test cases, including the invalid-choice case. Hence the algorithm is logically correct and handles the edge cases correctly.

Step 4: C Program

```
/* Menu-driven C program using switch-case to input a
   positive integer and perform one of the following
   operations based on the user's choice:
     1. Sum of squares of digits   -> while loop
     2. Sum of cubes of digits     -> for loop
     3. Difference of (sum of cubes - sum of squares) -> do-while loop
*/

#include <stdio.h>

int main()
{
    int num, choice, digit, n;
    int sumSquares = 0, sumCubes = 0, diff;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    printf("\n----- MENU -----\n");
    printf("1. Sum of squares of digits (while loop)\n");
    printf("2. Sum of cubes of digits (for loop)\n");
    printf("3. Difference between sum of cubes and sum of squares "
           "(do-while loop)\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
        {
            n = num;
            sumSquares = 0;
            while (n != 0)
            {
                digit = n % 10;
                sumSquares += digit * digit;
                n = n / 10;
            }
            printf("Sum of squares of digits = %d\n", sumSquares);
            break;
        }

        case 2:
        {
            sumCubes = 0;
```

```
            for (n = num; n != 0; n = n / 10)
            {
                digit = n % 10;
                sumCubes += digit * digit * digit;
            }
            printf("Sum of cubes of digits = %d\n", sumCubes);
            break;
        }

        case 3:
        {
            n = num;
            sumSquares = 0;
            sumCubes = 0;
            do
            {
                digit = n % 10;
                sumSquares += digit * digit;
                sumCubes += digit * digit * digit;
                n = n / 10;
            } while (n != 0);

            diff = sumCubes - sumSquares;
            printf("Difference (Sum of cubes - Sum of squares) = %d\n", diff);
            break;
        }

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}
```

Step 5: Compile, Execute, and Evaluate

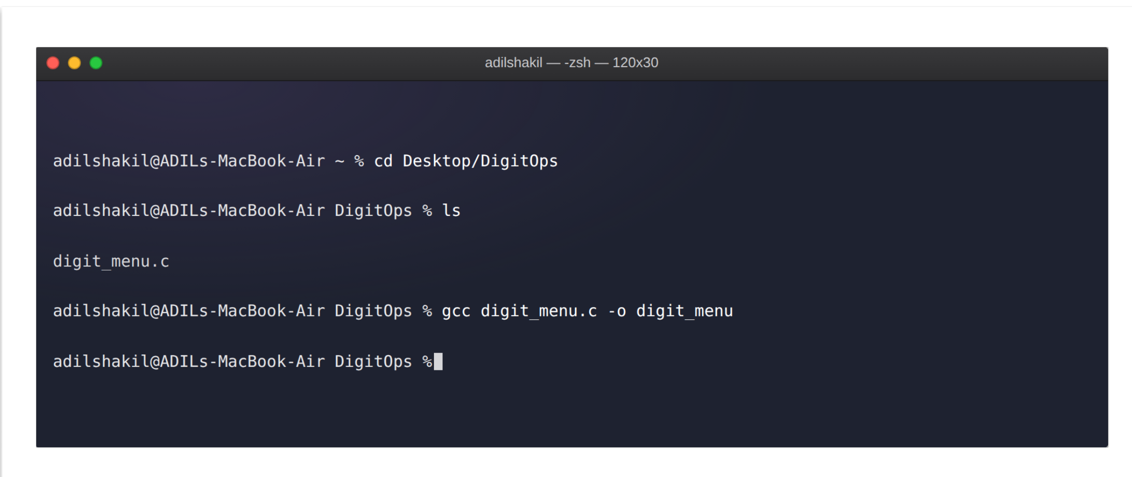
The program was compiled using the GCC compiler on macOS Terminal (zsh) with the command shown below. The compilation completed with no errors or warnings, and the executable was run once for each of the 6 test cases prepared in Step 2.

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	num = 123 choice = 1	Sum of squares of digits = 14	Sum of squares of digits = 14	**Pass**
2	num = 123 choice = 2	Sum of cubes of digits = 36	Sum of cubes of digits = 36	**Pass**

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
3	num = 123 choice = 3	Difference (Sum of cubes - Sum of squares) = 22	Difference (Sum of cubes - Sum of squares) = 22	**Pass**
4	num = 9 choice = 1	Sum of squares of digits = 81	Sum of squares of digits = 81	**Pass**
5	num = 45 choice = 3	Difference (Sum of cubes - Sum of squares) = 148	Difference (Sum of cubes - Sum of squares) = 148	**Pass**
6	num = 123 choice = 9	Invalid choice!	Invalid choice!	**Pass**

Actual output for all 6 test cases matches the expected output exactly, so every test case is marked Pass. This confirms that the while loop, for loop, and do-while loop branches of the switch-case all work correctly, and invalid menu choices are handled gracefully.

Step 6: Compilation and Execution Screenshots



```
adilshakil — -zsh — 120x30

adilshakil@ADILs-MacBook-Air DigitOps % ./digit_menu
Enter a positive integer: 123

----- MENU -----
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 1
Sum of squares of digits = 14
adilshakil@ADILs-MacBook-Air DigitOps %
```

```
adilshakil — -zsh — 120x30

adilshakil@ADILs-MacBook-Air DigitOps % ./digit_menu
Enter a positive integer: 123

----- MENU -----
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 2
Sum of cubes of digits = 36
adilshakil@ADILs-MacBook-Air DigitOps %
```

adilshakil — -zsh — 120x30

```
adilshakil@ADILs-MacBook-Air DigitOps % ./digit_menu
Enter a positive integer: 123

----- MENU -----
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 3
Difference (Sum of cubes - Sum of squares) = 22
adilshakil@ADILs-MacBook-Air DigitOps %
```

adilshakil — -zsh — 120x30

```
adilshakil@ADILs-MacBook-Air DigitOps % ./digit_menu
Enter a positive integer: 9

----- MENU -----
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 1
Sum of squares of digits = 81
adilshakil@ADILs-MacBook-Air DigitOps %
```

```
adilshakil — -zsh — 120x30

adilshakil@ADILs-MacBook-Air DigitOps % ./digit_menu
Enter a positive integer: 45

----- MENU -----
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 3
Difference (Sum of cubes - Sum of squares) = 148
adilshakil@ADILs-MacBook-Air DigitOps %
```

```
adilshakil — -zsh — 120x30

adilshakil@ADILs-MacBook-Air DigitOps % ./digit_menu
Enter a positive integer: 123

----- MENU -----
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 9
Invalid choice!
adilshakil@ADILs-MacBook-Air DigitOps %
```